

JOIcode Side-by-Side Report · gpt41_mini

Side-by-side GT vs Generated JOIcode report for **gpt41_mini**. This report is rendered from row-level benchmark outputs and is print-friendly for PDF export.

ROWS 23	AVG DET 67.6774	DET PASS 0.0%	GT EXACT 0.0%	AVG PROMPT TOKENS 18177.3
AVG COMPLETION TOKENS 68.7	AVG LLM LATENCY 2.695s	P50 LLM LATENCY 2.494s	PEAK VRAM 0.000 GB	OOM COUNT 0

RUN METADATA

Results Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_by_category_manual_20260506/retrieval_gpt41_mini_cat4
Output Dir	Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_by_category_manual_20260506/retrieval_gpt41_mini_cat4/side_by_side_report_gpt41_mini_failures_only
Suite	paper_with_cloud_ref
Genome	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/best_genome.json
Created At	2026-05-06T16:36:27.366326
Categories	4
Limit/Category	-
Rows	91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120

TOP FAILURE REASONS

dataflow	23
gt_mismatch	23
semantic	9
gt_receiver_coverage	6
gt_service_coverage	6
numeric_grounding	5
extraneous	4
service_match	3

CONTEXT + C

service_retrie
connected_d
Category 4

MODEL RUNTIME SNAPSHOT

MODEL GPT-4.1-mini	RESOLVED MODEL -	WORKER PYTHON -	SERVICE LIST service_list_ver2.0.1.json	CANDIDATE K 1
REPAIR ATTEMPTS 0	DET PROFILE openai			

ROW DETAILS

Row 91 · Category 4

When the contact sensor is detected, turn on the light.

접촉 센서가 감지되면 조명을 켜줘.

GT name: WhenContactSensorIsDetectedTurnOnLight				
Service list: service_list_ver2.0.1.json				
Output name: ContactSensorTriggerLightOn				
Retrieval cats: LightSensor, ContactSensor, Light, PresenceVitalSensor, PresenceSensor, MotionSensor, DimmerSwitch, CarbonDioxideSensor, TemperatureSensor, RainSensor				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.5323	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 19607
COMPLETION TOKENS 86	TOTAL TOKENS 19693	LLM LATENCY 3.738s	PIPELINE 9.257s	TOK/S 5268.4663
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
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LightSensor

1 services · 1 values · 0 functions

VALUES

LightSensor_Brightness

raw Brightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

None

ContactSensor

1 services · 1 values · 0 functions

VALUES

ContactSensor_Contact
raw Contact · arg - · ret BOOL · Allows reading the value of a contact sensor device

FUNCTIONS

None

Light
15 services · 8 values · 7 functions

RETRIEVED · RANK 3

VALUES
Light_ColorMode raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity
Light_CurrentBrightness raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity
Light_CurrentColorTemperature raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity
Light_CurrentHue raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity
Light_CurrentRGB raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity
Light_CurrentSaturation raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity
Light_CurrentX raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity
Light_CurrentY raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity
FUNCTIONS
Light_MoveToBrightness raw MoveToBrightness · arg DOUBLE DOUBLE · ret VOID · A numerical representation of the brightness intensity
Light_MoveToColorTemperature raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity
Light_MoveToHue raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity
Light_MoveToHueAndSaturation raw MoveToHueAndSaturation · arg DOUBLE DOUBLE · ret VOID · A numerical representation of the brightness intensity
Light_MoveToRGB raw MoveToRGB · arg INTEGER INTEGER INTEGER · ret VOID · A numerical representation of the brightness intensity
Light_MoveToSaturation raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity
Light_MoveToXY raw MoveToXY · arg DOUBLE DOUBLE · ret VOID · A numerical representation of the brightness intensity

PresenceVitalSensor 7 services · 7 values · 0 functions	RETRIEVED · RANK 4
VALUES	
PresenceVitalSensor_Awakeness raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection	

PresenceVitalSensor_Distance

raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_DwellTime

raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_HeartRate

raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_MovementIndex

raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Presence

raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_RespiratoryRate

raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS

None

PresenceSensor

RETRIEVED · RANK 5

1 services · 1 values · 0 functions

VALUES

PresenceSensor_Presence

raw Presence · arg - · ret BOOL · The ability to see the current status of a presence sensor device

FUNCTIONS

None

MotionSensor

RETRIEVED · RANK 6

1 services · 1 values · 0 functions

VALUES

MotionSensor_Motion

raw Motion · arg - · ret BOOL · Motion sensor device

FUNCTIONS

None

DimmerSwitch

RETRIEVED · RANK 7

4 services · 4 values · 0 functions

VALUES

DimmerSwitch_Button1

raw Button1 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button2

raw Button2 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button3

raw Button3 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button4

raw Button4 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

FUNCTIONS

None

CarbonDioxideSensor

RETRIEVED · RANK 8

1 services · 1 values · 0 functions

VALUES

CarbonDioxideSensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Measure carbon dioxide levels

FUNCTIONS

None

TemperatureSensor

RETRIEVED · RANK 9

1 services · 1 values · 0 functions

VALUES

TemperatureSensor_Temperature

raw Temperature · arg - · ret DOUBLE · Get the temperature from a Device that reports current temperature

FUNCTIONS

None

RainSensor

RETRIEVED · RANK 10

1 services · 1 values · 0 functions

VALUES

RainSensor_Rain

raw Rain · arg - · ret BOOL · A Device that senses rain

FUNCTIONS

None

GT JOICODE

cron="" · period=0

```
wait until ((#ContactSensor).contactsensor_contact == true)

(#Light).switch_on()
```

GENERATED JOICODE

cron="" · period=100

```
triggered := false
if ((#ContactSensor).contactsensor_contact == true) {
  if (triggered == false) {
    (#Light).switch_on()
    triggered = true
  }
} else {
  triggered = false
}
```

FAILURE REASONS

LINE DIFF

dataflow

gt_mismatch

GT	Generated
wait until ((#ContactSensor).contactsensor_contact == true)	triggered := false
	if ((#ContactSensor).contactsensor_contact == true) {
(#Light).switch_on()	if (triggered == false) {
	(#Light).switch_on()
	triggered = true
	}
	} else {
	triggered = false
	}

Row 92 · Category 4

When the presence sensor detects someone, turn on the TV.

재질 센서가 감지되면 TV를 켜줘.

GT name: WhenPresenceSensorDetectsSomeoneTurnOnTV				
Service list: service_list_ver2.0.1.json				
Output name: PresenceDetectedTurnOnTV				
Retrieval cats: PresenceSensor, Television, PresenceVitalSensor, TemperatureSensor, ContactSensor, MotionSensor, Switch, PressureSensor, CarbonDioxideSensor, LeakSensor				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.5436	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 18276
COMPLETION TOKENS 88	TOTAL TOKENS 18364	LLM LATENCY 2.457s	PIPELINE 3.367s	TOK/S 7473.5471
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
--------------------------------------	--	-------------------	------------------------	-----------------------	---------------------------

PresenceSensor

1 services · 1 values · 0 functions

VALUES

PresenceSensor_Presence

raw Presence · arg - · ret BOOL · The ability to see the current status of a presence sensor device

FUNCTIONS

None

Television

4 services · 1 values · 3 functions

VALUES

Television_Channel

raw Channel · arg - · ret INTEGER · A television device

FUNCTIONS

Television_ChannelDown

raw ChannelDown · arg - · ret INTEGER · A television device

Television_ChannelUp

raw ChannelUp · arg - · ret INTEGER · A television device

Television_SetChannel

raw SetChannel · arg INTEGER · ret VOID · A television device

PresenceVitalSensor

RETRIEVED · RANK 3

7 services · 7 values · 0 functions

VALUES

PresenceVitalSensor_Awakeness

raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Distance

raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_DwellTime

raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_HeartRate

raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_MovementIndex

raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Presence

raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_RespiratoryRate

raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS

None

TemperatureSensor

RETRIEVED · RANK 4

1 services · 1 values · 0 functions

VALUES

TemperatureSensor_Temperature

raw Temperature · arg - · ret DOUBLE · Get the temperature from a Device that reports current temperature

FUNCTIONS

None

ContactSensor

RETRIEVED · RANK 5

1 services · 1 values · 0 functions

VALUES

ContactSensor_Contact

raw Contact · arg - · ret BOOL · Allows reading the value of a contact sensor device

FUNCTIONS

None

MotionSensor

RETRIEVED · RANK 6

1 services · 1 values · 0 functions

VALUES

MotionSensor_Motion

raw Motion · arg - · ret BOOL · Motion sensor device

FUNCTIONS

None

Switch

RETRIEVED · RANK 7

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

PressureSensor

RETRIEVED · RANK 8

1 services · 1 values · 0 functions

VALUES

PressureSensor_Pressure

raw Pressure · arg - · ret DOUBLE · The ability to see the current status of a pressure sensor device

FUNCTIONS

None

CarbonDioxideSensor

RETRIEVED · RANK 9

1 services · 1 values · 0 functions

VALUES

CarbonDioxideSensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Measure carbon dioxide levels

FUNCTIONS

None

LeakSensor

1 services · 1 values · 0 functions

RETRIEVED · RANK 10

VALUES

LeakSensor_Leakage

raw Leakage · arg - · ret BOOL · A Device that senses water leakage

FUNCTIONS

None

GT JOICODE

cron="" · period=0

```
wait until ((#PresenceSensor).presencesensor_presence == true)

(#Television).switch_on()
```

GENERATED JOICODE

cron="" · period=100

```
triggered := false
if ((#PresenceSensor).presencesensor_presence == true) {
  if (triggered == false) {
    (#Television).switch_on()
    triggered = true
  }
} else {
  triggered = false
}
```

FAILURE REASONS

dataflow

gt_mismatch

semantic

LINE DIFF

GT	Generated
wait until ((#PresenceSensor).presencesensor_presence == true)	triggered := false
	if ((#PresenceSensor).presencesensor_presence == true) {
(#Television).switch_on()	if (triggered == false) {
	(#Television).switch_on()
	triggered = true
	}
	} else {
	triggered = false
	}

Row 97 · Category 4

When smoke is detected, sound the fire alarm with the siren.

연기가 감지되면 사이렌으로 화재경보를 울려줘.

GT name: WhenSmokeIsDetectedSoundFireAlarmWith				
Service list: service_list_ver2.0.1.json				
Output name: SmokeDetectedFireAlarm				
Retrieval cats: SmokeDetector, Siren, SoundSensor, AudioRecorder, AirPurifier, PresenceVitalSensor, AirQualitySensor, Oven, Safe, Door				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8186	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 19929
COMPLETION TOKENS 62	TOTAL TOKENS 19991	LLM LATENCY 2.272s	PIPELINE 3.239s	TOK/S 8797.6940
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
--------------------------------------	--	-------------------	------------------------	-----------------------	---------------------------

SmokeDetector

1 services · 1 values · 0 functions

VALUES

SmokeDetector_Smoke
raw Smoke · arg - · ret BOOL · A Device that detects smoke

FUNCTIONS

None

RETRIEVED · RANK 1

Siren

2 services · 1 values · 1 functions

RETRIEVED · RANK 2

VALUES

Siren_SirenMode
raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode
raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

SoundSensor

RETRIEVED · RANK 3

1 services · 1 values · 0 functions

VALUES

SoundSensor_Sound
raw Sound · arg - · ret DOUBLE · Sound sensor device for measuring sound levels

FUNCTIONS

None

AudioRecorder

RETRIEVED · RANK 4

5 services · 2 values · 3 functions

VALUES

AudioRecorder_AudioFile
raw AudioFile · arg - · ret BINARY · Record audio

AudioRecorder_RecordStatus
raw RecordStatus · arg - · ret ENUM · Record audio

FUNCTIONS

AudioRecorder_RecordStart
raw RecordStart · arg - · ret VOID · Record audio

AudioRecorder_RecordStop
raw RecordStop · arg BINARY · ret VOID · Record audio

AudioRecorder_RecordWithDuration
raw RecordWithDuration · arg STRING | DOUBLE · ret BINARY · Record audio

AirPurifier

RETRIEVED · RANK 5

2 services · 1 values · 1 functions

VALUES

AirPurifier_AirPurifierMode
raw AirPurifierMode · arg - · ret ENUM · Controls air purifier mode

FUNCTIONS

AirPurifier_SetAirPurifierMode
raw SetAirPurifierMode · arg ENUM · ret VOID · Controls air purifier mode

PresenceVitalSensor

RETRIEVED · RANK 6

7 services · 7 values · 0 functions

VALUES

PresenceVitalSensor_Awakeness

raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Distance

raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_DwellTime

raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_HeartRate

raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_MovementIndex

raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Presence

raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_RespiratoryRate

raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS

None

AirQualitySensor

RETRIEVED · RANK 7

7 services · 7 values · 0 functions

VALUES

AirQualitySensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_DustLevel

raw DustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_FineDustLevel

raw FineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Temperature

raw Temperature · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_TvocLevel

raw TvocLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_VeryFineDustLevel

raw VeryFineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

FUNCTIONS

None

Oven

RETRIEVED · RANK 8

4 services · 1 values · 3 functions

VALUES

Oven_OvenMode

raw OvenMode · arg - · ret ENUM · Allows for the control of the oven mode

FUNCTIONS

Oven_AddMoreTime

raw AddMoreTime · arg DOUBLE · ret VOID · Allows for the control of the oven mode

Oven_SetCookingParameters

raw SetCookingParameters · arg ENUM | DOUBLE · ret VOID · Allows for the control of the oven mode

Oven_SetOvenMode

raw SetOvenMode · arg ENUM · ret VOID · Allows for the control of the oven mode

Safe

RETRIEVED · RANK 9

3 services · 1 values · 2 functions

VALUES

Safe_SafeState

raw SafeState · arg - · ret ENUM · Allows for the control of the Safe

FUNCTIONS

Safe_Lock

raw Lock · arg - · ret VOID · Allows for the control of the Safe

Safe_Unlock

raw Unlock · arg - · ret VOID · Allows for the control of the Safe

Door

RETRIEVED · RANK 10

3 services · 1 values · 2 functions

VALUES

Door_DoorState

raw DoorState · arg - · ret ENUM · Allow for the control of a door

FUNCTIONS

Door_Close

raw Close · arg - · ret VOID · Allow for the control of a door

Door_Open

raw Open · arg - · ret VOID · Allow for the control of a door

GT JOICODE

cron="" · period=0

```
wait until ((#SmokeDetector).smokedetector_smoke == true)

(#Siren).siren_setsirenmode("fire")
```

GENERATED JOICODE

cron="" · period=0

```
if ((#SmokeDetector).smokedetector_smoke == true) {
  (#Siren).siren_setsirenmode("fire")
}
```

FAILURE REASONS

dataflow gt_mismatch

LINE DIFF

GT	Generated
wait until ((#SmokeDetector).smokedetector_smoke == true)	if ((#SmokeDetector).smokedetector_smoke == true) {
	(#Siren).siren_setsirenmode("fire")
(#Siren).siren_setsirenmode("fire")	}

Row 99 · Category 4

When the temperature reaches 35 degrees or higher, set the air conditioner to 30 degrees.

온도가 35도 이상이 되면 에어컨의 목표 온도를 30도로 설정해줘.

GT name: WhenTemperatureReaches35DegreesOrHigherSet				
Service list: service_list_ver2.0.1.json				
Output name: SetAirConditionerTo30WhenTempAbove35				
Retrieval cats: AirConditioner, AirPurifier, AirQualitySensor, TemperatureSensor, LaundryDryer, Dishwasher, WeatherProvider, Dehumidifier, ArmRobot, RiceCooker				
DET 58.7426	PASS No	GT EXACT No	SIMILARITY 0.8485	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.0000	NUMERIC 0.0000	ENUM 1.0000	PROMPT TOKENS 20520
COMPLETION TOKENS 68	TOTAL TOKENS 20588	LLM LATENCY 2.494s	PIPELINE 3.461s	TOK/S 8256.0051
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
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AirConditioner

4 services · 2 values · 2 functions

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

AirPurifier

RETRIEVED · RANK 2

2 services · 1 values · 1 functions

VALUES

AirPurifier_AirPurifierMode

raw AirPurifierMode · arg - · ret ENUM · Controls air purifier mode

FUNCTIONS

AirPurifier_SetAirPurifierMode

raw SetAirPurifierMode · arg ENUM · ret VOID · Controls air purifier mode

AirQualitySensor

RETRIEVED · RANK 3

7 services · 7 values · 0 functions

VALUES

AirQualitySensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_DustLevel

raw DustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_FineDustLevel

raw FineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Temperature

raw Temperature · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_TvocLevel

raw TvocLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_VeryFineDustLevel

raw VeryFineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

FUNCTIONS

None

TemperatureSensor

RETRIEVED · RANK 4

1 services · 1 values · 0 functions

VALUES

TemperatureSensor_Temperature

raw Temperature · arg - · ret DOUBLE · Get the temperature from a Device that reports current temperature

FUNCTIONS

None

LaundryDryer

4 services · 2 values · 2 functions

RETRIEVED · RANK 5

VALUES

LaundryDryer_LaundryDryerMode

raw LaundryDryerMode · arg - · ret ENUM · Allows for the control of the laundry dryer mode

LaundryDryer_SpinSpeed

raw SpinSpeed · arg - · ret INTEGER · Allows for the control of the laundry dryer mode

FUNCTIONS

LaundryDryer_SetLaundryDryerMode

raw SetLaundryDryerMode · arg ENUM · ret VOID · Allows for the control of the laundry dryer mode

LaundryDryer_SetSpinSpeed

raw SetSpinSpeed · arg INTEGER · ret VOID · Allows for the control of the laundry dryer mode

Dishwasher

2 services · 1 values · 1 functions

RETRIEVED · RANK 6

VALUES

Dishwasher_DishwasherMode

raw DishwasherMode · arg - · ret ENUM · Allows for the control of the dishwasher mode

FUNCTIONS

Dishwasher_SetDishwasherMode

raw SetDishwasherMode · arg ENUM · ret VOID · Allows for the control of the dishwasher mode

WeatherProvider

7 services · 6 values · 1 functions

RETRIEVED · RANK 7

VALUES

WeatherProvider_HumidityWeather

raw HumidityWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Pm10Weather

raw Pm10Weather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Pm25Weather

raw Pm25Weather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_PressureWeather

raw PressureWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_TemperatureWeather

raw TemperatureWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Weather

raw Weather · arg - · ret ENUM · Provides weather information

FUNCTIONS

file:///home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_by_category_manual_20260506/retrieval_gpt41_mini_cat4/side_by_side_report...

19/79

WeatherProvider_GetWeatherInfo
raw GetWeatherInfo · arg DOUBLE | DOUBLE · ret STRING · Provides weather information

Dehumidifier

RETRIEVED · RANK 8

2 services · 1 values · 1 functions

VALUES

Dehumidifier_DehumidifierMode
raw DehumidifierMode · arg - · ret ENUM · Allows for the control of the dehumidifier mode

FUNCTIONS

Dehumidifier_SetDehumidifierMode
raw SetDehumidifierMode · arg ENUM · ret VOID · Allows for the control of the dehumidifier mode

ArmRobot

RETRIEVED · RANK 9

5 services · 2 values · 3 functions

VALUES

ArmRobot_ArmRobotType
raw ArmRobotType · arg - · ret ENUM · Allows for the control of the arm robot

ArmRobot_CurrentPosition
raw CurrentPosition · arg - · ret STRING · Allows for the control of the arm robot

FUNCTIONS

ArmRobot_Hello
raw Hello · arg - · ret VOID · Allows for the control of the arm robot

ArmRobot_SendCommand
raw SendCommand · arg ENUM · ret VOID · Allows for the control of the arm robot

ArmRobot_SetPosition
raw SetPosition · arg ENUM · ret VOID · Allows for the control of the arm robot

RiceCooker

RETRIEVED · RANK 10

4 services · 1 values · 3 functions

VALUES

RiceCooker_RiceCookerMode
raw RiceCookerMode · arg - · ret ENUM · Allows for the control of the Rice Cooker

FUNCTIONS

RiceCooker_AddMoreTime
raw AddMoreTime · arg DOUBLE · ret VOID · Allows for the control of the Rice Cooker

RiceCooker_SetCookingParameters
raw SetCookingParameters · arg ENUM | DOUBLE · ret VOID · Allows for the control of the Rice Cooker

RiceCooker_SetRiceCookerMode
raw SetRiceCookerMode · arg ENUM · ret VOID · Allows for the control of the Rice Cooker

GT JOICODE

cron="" · period=0

```
wait until ((#TemperatureSensor).temperaturesensor_temperature >= 35)

(#AirConditioner).airconditioner_settargettemperature(30)
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#TemperatureSensor).temperaturemeasurement_temperature >= 35.0)
(#AirConditioner).airconditioner_settargettemperature(30.0)
```

FAILURE REASONS

- dataflow
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- numeric_grounding
- service_match
- unknown_service:temperaturemeasurement_temperature

LINE DIFF

GT	Generated
wait until ((#TemperatureSensor).temperaturesensor_temperature >= 35)	wait until ((#TemperatureSensor).temperaturemeasurement_temperature >= 35.0)
	(#AirConditioner).airconditioner_settargettemperature(30.0)
(#AirConditioner).airconditioner_settargettemperature(30)	

Row 100 · Category 4

When the safe is unlocked, sound the police siren.

금고가 열리면 경찰 사이렌을 울려줘.

GT name: WhenSafelsUnlockedSoundPoliceSiren				
Service list: service_list_ver2.0.1.json				
Output name: SafeUnlockPoliceSiren				
Retrieval cats: Safe, Siren, SoundSensor, DoorLock, PresenceVitalSensor, Door, AudioRecorder, Speaker, Plug, TapDialSwitch				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9778	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 20413
COMPLETION TOKENS 61	TOTAL TOKENS 20474	LLM LATENCY 2.414s	PIPELINE 3.382s	TOK/S 8480.6561
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

VALUES	
Safe_SafeState raw SafeState · arg - · ret ENUM · Allows for the control of the Safe	
FUNCTIONS	
Safe_Lock raw Lock · arg - · ret VOID · Allows for the control of the Safe	
Safe_Unlock raw Unlock · arg - · ret VOID · Allows for the control of the Safe	
Siren	RETRIEVED · RANK 2
2 services · 1 values · 1 functions	
VALUES	
Siren_SirenMode raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren	
FUNCTIONS	
Siren_SetSirenMode raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren	
SoundSensor	RETRIEVED · RANK 3
1 services · 1 values · 0 functions	
VALUES	
SoundSensor_Sound raw Sound · arg - · ret DOUBLE · Sound sensor device for measuring sound levels	
FUNCTIONS	
None	
DoorLock	RETRIEVED · RANK 4
3 services · 1 values · 2 functions	
VALUES	
DoorLock_DoorLockState raw DoorLockState · arg - · ret ENUM · Allow for the control of a door lock	
FUNCTIONS	
DoorLock_Lock raw Lock · arg - · ret VOID · Allow for the control of a door lock	
DoorLock_Unlock raw Unlock · arg - · ret VOID · Allow for the control of a door lock	
PresenceVitalSensor	RETRIEVED · RANK 5
7 services · 7 values · 0 functions	
VALUES	

- PresenceVitalSensor_Awakeness**
raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection
- PresenceVitalSensor_Distance**
raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection
- PresenceVitalSensor_DwellTime**
raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection
- PresenceVitalSensor_HeartRate**
raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection
- PresenceVitalSensor_MovementIndex**
raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection
- PresenceVitalSensor_Presence**
raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection
- PresenceVitalSensor_RespiratoryRate**
raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS

None

Door

RETRIEVED · RANK 6

3 services · 1 values · 2 functions

VALUES

- Door_DoorState**
raw DoorState · arg - · ret ENUM · Allow for the control of a door

FUNCTIONS

- Door_Close**
raw Close · arg - · ret VOID · Allow for the control of a door
- Door_Open**
raw Open · arg - · ret VOID · Allow for the control of a door

AudioRecorder

RETRIEVED · RANK 7

5 services · 2 values · 3 functions

VALUES

- AudioRecorder_AudioFile**
raw AudioFile · arg - · ret BINARY · Record audio
- AudioRecorder_RecordStatus**
raw RecordStatus · arg - · ret ENUM · Record audio

FUNCTIONS

- AudioRecorder_RecordStart**
raw RecordStart · arg - · ret VOID · Record audio
- AudioRecorder_RecordStop**
raw RecordStop · arg BINARY · ret VOID · Record audio

AudioRecorder_RecordWithDuration
raw RecordWithDuration · arg STRING | DOUBLE · ret BINARY · Record audio

Speaker

RETRIEVED · RANK 8

11 services · 2 values · 9 functions

VALUES

Speaker_PlaybackState
raw PlaybackState · arg - · ret ENUM · Speaker device for audio playback and media control

Speaker_Volume
raw Volume · arg - · ret INTEGER · Speaker device for audio playback and media control

FUNCTIONS

Speaker_FastForward
raw FastForward · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Pause
raw Pause · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Play
raw Play · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Rewind
raw Rewind · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_SetVolume
raw SetVolume · arg INTEGER · ret INTEGER · Speaker device for audio playback and media control

Speaker_Speak
raw Speak · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Stop
raw Stop · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_VolumeDown
raw VolumeDown · arg - · ret INTEGER · Speaker device for audio playback and media control

Speaker_VolumeUp
raw VolumeUp · arg - · ret INTEGER · Speaker device for audio playback and media control

Plug

RETRIEVED · RANK 9

3 services · 3 values · 0 functions

VALUES

Plug_Current
raw Current · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Power
raw Power · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Voltage
raw Voltage · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

FUNCTIONS

None

TapDialSwitch

6 services · 6 values · 0 functions

RETRIEVED · RANK 10

VALUES

TapDialSwitch_Button1

raw Button1 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Button2

raw Button2 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Button3

raw Button3 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Button4

raw Button4 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Rotation

raw Rotation · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_RotationSteps

raw RotationSteps · arg - · ret INTEGER · A tap dial switch device with multiple buttons and a rotary dial

FUNCTIONS

None

GT JOICODE

cron="" · period=0

```
wait until ((#Safe).safe_safestate == "unlocked")

(#Siren).siren_setsirenmode("police")
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#Safe).safe_safestate == "open")
(#Siren).siren_setsirenmode("police")
```

FAILURE REASONS

- dataflow
- gt_mismatch

LINE DIFF

GT	Generated
wait until ((#Safe).safe_safestate == "unlocked")	wait until ((#Safe).safe_safestate == "open")
	(#Siren).siren_setsirenmode("police")
(#Siren).siren_setsirenmode("police")	

Row 102 · Category 4

When the humidity exceeds 80%, start the dehumidifier in drying mode.

습도가 80%보다 커지면 제습기를 건조 모드로 작동시켜줘.

GT name: WhenHumidityExceeds80StartDehumidifierInDrying				
Service list: service_list_ver2.0.1.json				
Output name: StartDehumidifierWhenHumidityExceeds80				
Retrieval cats: Dehumidifier, Humidifier, HumiditySensor, Dishwasher, LaundryDryer, AirPurifier, AirQualitySensor, AirConditioner, WeatherProvider, RobotVacuumCleaner				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9961	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 0.0000	ENUM 1.0000	PROMPT TOKENS 20074
COMPLETION TOKENS 70	TOTAL TOKENS 20144	LLM LATENCY 1.806s	PIPELINE 2.768s	TOK/S 11154.5490
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

Humidifier

RETRIEVED · RANK 2

2 services · 1 values · 1 functions

VALUES

Humidifier_HumidifierMode

raw HumidifierMode · arg - · ret ENUM · Maintains and sets the state of an humidifier

FUNCTIONS

Humidifier_SetHumidifierMode

raw SetHumidifierMode · arg ENUM · ret VOID · Maintains and sets the state of an humidifier

HumiditySensor

RETRIEVED · RANK 3

1 services · 1 values · 0 functions

VALUES

HumiditySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Allow reading the relative humidity from devices that support it

FUNCTIONS

None

Dishwasher

RETRIEVED · RANK 4

2 services · 1 values · 1 functions

VALUES

Dishwasher_DishwasherMode

raw DishwasherMode · arg - · ret ENUM · Allows for the control of the dishwasher mode

FUNCTIONS

Dishwasher_SetDishwasherMode

raw SetDishwasherMode · arg ENUM · ret VOID · Allows for the control of the dishwasher mode

LaundryDryer

RETRIEVED · RANK 5

4 services · 2 values · 2 functions

VALUES

LaundryDryer_LaundryDryerMode

raw LaundryDryerMode · arg - · ret ENUM · Allows for the control of the laundry dryer mode

LaundryDryer_SpinSpeed

raw SpinSpeed · arg - · ret INTEGER · Allows for the control of the laundry dryer mode

FUNCTIONS

LaundryDryer_SetLaundryDryerMode

raw SetLaundryDryerMode · arg ENUM · ret VOID · Allows for the control of the laundry dryer mode

LaundryDryer_SetSpinSpeed

raw SetSpinSpeed · arg INTEGER · ret VOID · Allows for the control of the laundry dryer mode

AirPurifier

RETRIEVED · RANK 6

2 services · 1 values · 1 functions

VALUES

AirPurifier_AirPurifierMode

raw AirPurifierMode · arg - · ret ENUM · Controls air purifier mode

FUNCTIONS

AirPurifier_SetAirPurifierMode

raw SetAirPurifierMode · arg ENUM · ret VOID · Controls air purifier mode

AirQualitySensor

RETRIEVED · RANK 7

7 services · 7 values · 0 functions

VALUES

AirQualitySensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_DustLevel

raw DustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_FineDustLevel

raw FineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Temperature

raw Temperature · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_TvocLevel

raw TvocLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_VeryFineDustLevel

raw VeryFineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

FUNCTIONS

None

AirConditioner

RETRIEVED · RANK 8

4 services · 2 values · 2 functions

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

WeatherProvider

RETRIEVED · RANK 9

7 services · 6 values · 1 functions

VALUES

WeatherProvider_HumidityWeather

raw HumidityWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Pm10Weather

raw Pm10Weather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Pm25Weather

raw Pm25Weather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_PressureWeather

raw PressureWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_TemperatureWeather

raw TemperatureWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Weather

raw Weather · arg - · ret ENUM · Provides weather information

FUNCTIONS

WeatherProvider_GetWeatherInfo

raw GetWeatherInfo · arg DOUBLE | DOUBLE · ret STRING · Provides weather information

RobotVacuumCleaner

RETRIEVED · RANK 10

2 services · 1 values · 1 functions

VALUES

RobotVacuumCleaner_RobotVacuumCleanerMode

raw RobotVacuumCleanerMode · arg - · ret ENUM · Allows for the control of the robot cleaner cleaning mode

FUNCTIONS

RobotVacuumCleaner_SetRobotVacuumCleanerModeMode

raw SetRobotVacuumCleanerModeMode · arg ENUM · ret VOID · Allows for the control of the robot cleaner cleaning mode

GT JOICODE

cron="" · period=0

wait until ((#HumiditySensor).humiditysensor_humidity > 80)

(#Dehumidifier).dehumidifier_setdehumidifiermode("drying")

GENERATED JOICODE

cron="" · period=0

wait until ((#HumiditySensor).humiditysensor_humidity > 80.0)
(#Dehumidifier).dehumidifier_setdehumidifiermode("drying")

FAILURE REASONS

dataflow

gt_mismatch

numeric_grounding

LINE DIFF

GT	Generated
wait until ((#HumiditySensor).humiditysensor_humidity > 80)	wait until ((#HumiditySensor).humiditysensor_humidity > 80.0)
	(#Dehumidifier).dehumidifier_setdehumidifiermode("drying")

GT	Generated
(#Dehumidifier).dehumidifier_setdehumidifiermode("drying")	

Row 103 · Category 4

When motion is detected, turn on the light.

움직임이 감지되면 조명을 켜줘.

GT name: WhenMotionIsDetectedTurnOnLight				
Service list: service_list_ver2.0.1.json				
Output name: MotionDetectedTurnOnLight				
Retrieval cats: LightSensor, Light, MotionSensor, DimmerSwitch, Switch, PresenceVitalSensor, SmokeDetector, ColorControl, Television, Camera				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.5276	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 20770
COMPLETION TOKENS 86	TOTAL TOKENS 20856	LLM LATENCY 5.802s	PIPELINE 6.688s	TOK/S 3594.7465
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
--------------------------------------	--	-------------------	------------------------	-----------------------	---------------------------

LightSensor 1 services · 1 values · 0 functions VALUES LightSensor_Brightness raw Brightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity FUNCTIONS None	RETRIEVED · RANK 1
Light 15 services · 8 values · 7 functions	RETRIEVED · RANK 2

VALUES
Light_ColorMode raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity
Light_CurrentBrightness raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity
Light_CurrentColorTemperature raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity
Light_CurrentHue raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity
Light_CurrentRGB raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity
Light_CurrentSaturation raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity
Light_CurrentX raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity
Light_CurrentY raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity
FUNCTIONS
Light_MoveToBrightness raw MoveToBrightness · arg DOUBLE DOUBLE · ret VOID · A numerical representation of the brightness intensity
Light_MoveToColorTemperature raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity
Light_MoveToHue raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity
Light_MoveToHueAndSaturation raw MoveToHueAndSaturation · arg DOUBLE DOUBLE · ret VOID · A numerical representation of the brightness intensity
Light_MoveToRGB raw MoveToRGB · arg INTEGER INTEGER INTEGER · ret VOID · A numerical representation of the brightness intensity
Light_MoveToSaturation raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity
Light_MoveToXY raw MoveToXY · arg DOUBLE DOUBLE · ret VOID · A numerical representation of the brightness intensity

MotionSensor

RETRIEVED · RANK 3

1 services · 1 values · 0 functions

VALUES

MotionSensor_Motion
raw Motion · arg - · ret BOOL · Motion sensor device

FUNCTIONS

None

DimmerSwitch

RETRIEVED · RANK 4

4 services · 4 values · 0 functions

VALUES

DimmerSwitch_Button1
raw Button1 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button2
raw Button2 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button3
raw Button3 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button4
raw Button4 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

FUNCTIONS

None

Switch

RETRIEVED · RANK 5

4 services · 1 values · 3 functions

VALUES

Switch_Switch
raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off
raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On
raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle
raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

PresenceVitalSensor

RETRIEVED · RANK 6

7 services · 7 values · 0 functions

VALUES

PresenceVitalSensor_Awakeness
raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Distance
raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_DwellTime
raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_HeartRate
raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_MovementIndex
raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Presence
raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_RespiratoryRate
raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS
None

SmokeDetector RETRIEVED · RANK 7
1 services · 1 values · 0 functions

VALUES
SmokeDetector_Smoke
raw Smoke · arg - · ret BOOL · A Device that detects smoke

FUNCTIONS
None

ColorControl RETRIEVED · RANK 8
2 services · 1 values · 1 functions

ColorControl_Color
raw Color · arg - · ret STRING · Allows for control of a color changing device

FUNCTIONS
ColorControl_SetColor
raw SetColor · arg STRING · ret VOID · Allows for control of a color changing device

Television RETRIEVED · RANK 9
4 services · 1 values · 3 functions

Television_Channel
raw Channel · arg - · ret INTEGER · A television device

Television_ChannelDown
raw ChannelDown · arg - · ret INTEGER · A television device

Television_ChannelUp
raw ChannelUp · arg - · ret INTEGER · A television device

Television_SetChannel

raw SetChannel · arg INTEGER · ret VOID · A television device

Camera

RETRIEVED · RANK 10

8 services · 4 values · 4 functions

VALUES

Camera_CameraState

raw CameraState · arg - · ret ENUM · Controls camera device for image/video capture and streaming

Camera_Image

raw Image · arg - · ret BINARY · Controls camera device for image/video capture and streaming

Camera_Stream

raw Stream · arg - · ret STRING · Controls camera device for image/video capture and streaming

Camera_Video

raw Video · arg - · ret BINARY · Controls camera device for image/video capture and streaming

FUNCTIONS

Camera_CaptureImage

raw CaptureImage · arg - · ret BINARY · Controls camera device for image/video capture and streaming

Camera_CaptureVideo

raw CaptureVideo · arg - · ret BINARY · Controls camera device for image/video capture and streaming

Camera_StartStream

raw StartStream · arg - · ret STRING · Controls camera device for image/video capture and streaming

Camera_StopStream

raw StopStream · arg - · ret VOID · Controls camera device for image/video capture and streaming

GT JOICODE

cron="" · period=0

```
wait until ((#MotionSensor).motionsensor_motion == true)

(#Light).switch_on()
```

GENERATED JOICODE

cron="" · period=100

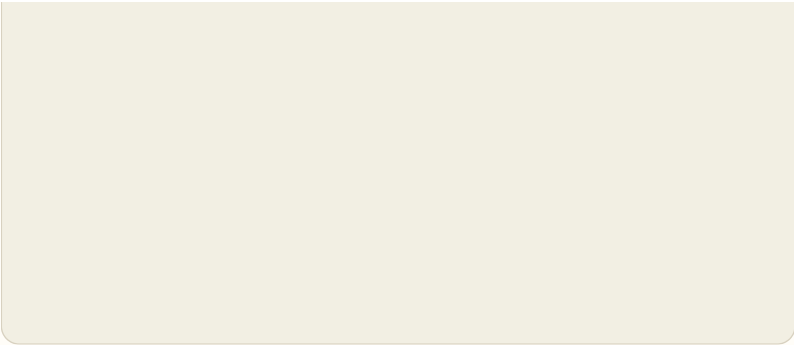
```
triggered := false
if ((#MotionSensor).motionsensor_motion == true) {
  if (triggered == false) {
    (#Light).switch_on()
    triggered = true
  }
} else {
  triggered = false
}
```

FAILURE REASONS

- dataflow
- gt_mismatch
- semantic

LINE DIFF

GT	Generated
wait until ((#MotionSensor).motionsensor_motion == true)	triggered := false
	if ((#MotionSensor).motionsensor_motion == true) {



GT	Generated
(#Light).switch_on()	if (triggered == false) {
	(#Light).switch_on()
	triggered = true
	}
	} else {
	triggered = false
	}

Row 105 · Category 4

When the charging voltage exceeds 250V, stop charging.

충전 전압이 250V보다 높아지면 충전을 중단해줘.

GT name: WhenChargingVoltageExceeds250VStopCharging				
Service list: service_list_ver2.0.1.json				
Output name: StopChargingWhenVoltageExceeds250V				
Retrieval cats: Charger, Plug, CarbonDioxideSensor, PresenceVitalSensor, Safe, AudioRecorder, Pump, RobotVacuumCleaner, PressureSensor, ArmRobot				
DET 61.2728	PASS No	GT EXACT No	SIMILARITY 0.8146	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.2000	NUMERIC 0.0000	ENUM 1.0000	PROMPT TOKENS 19511
COMPLETION TOKENS 61	TOTAL TOKENS 19572	LLM LATENCY 1.938s	PIPELINE 2.898s	TOK/S 10096.4663
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
--------------------------------------	--	-------------------	------------------------	-----------------------	---------------------------

Charger

4 services · 4 values · 0 functions

VALUES

Charger_ChargingState

raw ChargingState · arg - · ret ENUM · The current status of battery charging

Charger_Current

raw Current · arg - · ret DOUBLE · The current status of battery charging

Charger_Power

raw Power · arg - · ret DOUBLE · The current status of battery charging

Charger_Voltage

raw Voltage · arg - · ret DOUBLE · The current status of battery charging

FUNCTIONS

None

Plug

RETRIEVED · RANK 2

3 services · 3 values · 0 functions

VALUES

Plug_Current

raw Current · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Power

raw Power · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Voltage

raw Voltage · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

FUNCTIONS

None

CarbonDioxideSensor

RETRIEVED · RANK 3

1 services · 1 values · 0 functions

VALUES

CarbonDioxideSensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Measure carbon dioxide levels

FUNCTIONS

None

PresenceVitalSensor

RETRIEVED · RANK 4

7 services · 7 values · 0 functions

VALUES

PresenceVitalSensor_Awakeness

raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Distance

raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_DwellTime

raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_HeartRate

raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_MovementIndex

raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Presence

raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_RespiratoryRate

raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS

None

Safe

RETRIEVED · RANK 5

3 services · 1 values · 2 functions

VALUES

Safe_SafeState

raw SafeState · arg - · ret ENUM · Allows for the control of the Safe

FUNCTIONS

Safe_Lock

raw Lock · arg - · ret VOID · Allows for the control of the Safe

Safe_Unlock

raw Unlock · arg - · ret VOID · Allows for the control of the Safe

AudioRecorder

RETRIEVED · RANK 6

5 services · 2 values · 3 functions

VALUES

AudioRecorder_AudioFile

raw AudioFile · arg - · ret BINARY · Record audio

AudioRecorder_RecordStatus

raw RecordStatus · arg - · ret ENUM · Record audio

FUNCTIONS

AudioRecorder_RecordStart

raw RecordStart · arg - · ret VOID · Record audio

AudioRecorder_RecordStop

raw RecordStop · arg BINARY · ret VOID · Record audio

AudioRecorder_RecordWithDuration

raw RecordWithDuration · arg STRING | DOUBLE · ret BINARY · Record audio

Pump

RETRIEVED · RANK 7

2 services · 1 values · 1 functions

VALUES

Pump_PumpMode

raw PumpMode · arg - · ret ENUM · Allows for the control of a pump device

FUNCTIONS

Pump_SetPumpMode

raw SetPumpMode · arg ENUM · ret VOID · Allows for the control of a pump device

RobotVacuumCleaner

RETRIEVED · RANK 8

2 services · 1 values · 1 functions

VALUES

RobotVacuumCleaner_RobotVacuumCleanerMode

raw RobotVacuumCleanerMode · arg - · ret ENUM · Allows for the control of the robot cleaner cleaning mode

FUNCTIONS

RobotVacuumCleaner_SetRobotVacuumCleanerModeMode

raw SetRobotVacuumCleanerModeMode · arg ENUM · ret VOID · Allows for the control of the robot cleaner cleaning mode

PressureSensor

RETRIEVED · RANK 9

1 services · 1 values · 0 functions

VALUES

PressureSensor_Pressure

raw Pressure · arg - · ret DOUBLE · The ability to see the current status of a pressure sensor device

FUNCTIONS

None

ArmRobot

RETRIEVED · RANK 10

5 services · 2 values · 3 functions

VALUES

ArmRobot_ArmRobotType

raw ArmRobotType · arg - · ret ENUM · Allows for the control of the arm robot

ArmRobot_CurrentPosition

raw CurrentPosition · arg - · ret STRING · Allows for the control of the arm robot

FUNCTIONS

ArmRobot_Hello

raw Hello · arg - · ret VOID · Allows for the control of the arm robot

ArmRobot_SendCommand

raw SendCommand · arg ENUM · ret VOID · Allows for the control of the arm robot

ArmRobot_SetPosition

raw SetPosition · arg ENUM · ret VOID · Allows for the control of the arm robot

GT JOICODE

cron="" · period=0

```
wait until ((#Charger).charger_voltage > 250000)

(#Charger).switch_off()
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#Charger).charger_voltage > 250.0)
(#Charger).safe_lock()
```

FAILURE REASONS

- dataflow
- extraneous
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- numeric_grounding
- semantic

LINE DIFF

GT	Generated
wait until ((#Charger).charger_voltage > 250000)	wait until ((#Charger).charger_voltage > 250.0)
	(#Charger).safe_lock()
(#Charger).switch_off()	

Row 106 · Category 4

When the entrance door lock is unlocked, turn on the living room light.

현관 도어락이 잠금해제되면 거실 조명을 켜줘.

GT name: WhenEntranceDoorLockIsUnlockedTurnOn				
Service list: service_list_ver2.0.1.json				
Output name: EntranceDoorUnlockTurnOnLivingRoomLight				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9224	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 18391
COMPLETION TOKENS 59	TOTAL TOKENS 18450	LLM LATENCY 1.940s	PIPELINE 1.949s	TOK/S 9508.8388
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT					
Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.					
CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3

DoorLock	CONNECTED
3 services · 1 values · 2 functions	
VALUES	
DoorLock_DoorLockState raw DoorLockState · arg - · ret ENUM · Allow for the control of a door lock	
FUNCTIONS	
DoorLock_Lock raw Lock · arg - · ret VOID · Allow for the control of a door lock	
DoorLock_Unlock	

raw Unlock · arg - · ret VOID · Allow for the control of a door lock

Light

CONNECTED

15 services · 8 values · 7 functions

VALUES

Light_ColorMode
raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness
raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature
raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue
raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB
raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation
raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX
raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY
raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness
raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature
raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue
raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation
raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB
raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation
raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY
raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

GT JOICODE

cron="" · period=0

```
wait until ((#Entrance #DoorLock).doorlock_doorlockstate == "unlocked")

(#LivingRoom #Light).switch_on()
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#DoorLock).doorlock_doorlockstate == "open")
(#Light).switch_on()
```

FAILURE REASONS

- dataflow
- gt_mismatch
- semantic

LINE DIFF

GT	Generated
wait until ((#Entrance #DoorLock).doorlock_doorlockstate == "unlocked")	wait until ((#DoorLock).doorlock_doorlockstate == "open")
	(#Light).switch_on()
(#LivingRoom #Light).switch_on()	

Row 107 · Category 4

When motion is detected in the bedroom, turn on the bedroom air conditioner.

침실에 움직임이 감지되면 침실 에어컨을 켜줘.

GT name: WhenMotionIsDetectedInBedroomTurnOn				
Service list: service_list_ver2.0.1.json				
Output name: BedroomMotionACOn				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9391	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16692
COMPLETION TOKENS 67	TOTAL TOKENS 16759	LLM LATENCY 2.898s	PIPELINE 2.905s	TOK/S 5783.5525
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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MotionSensor

1 services · 1 values · 0 functions

VALUES

MotionSensor_Motion

raw Motion · arg - · ret BOOL · Motion sensor device

FUNCTIONS

None

AirConditioner

4 services · 2 values · 2 functions

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

GT JOICODE

cron="" · period=0

```
wait until ((#Bedroom #MotionSensor).motionsensor_motion == true)

(#Bedroom #AirConditioner).airconditioner_setairconditionermode("auto")
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#MotionSensor #bedroom).motionsensor_motion == true)
(#AirConditioner #bedroom).airconditioner_setairconditionermode("auto")
```

FAILURE REASONS

dataflow

gt_mismatch

LINE DIFF

GT	Generated
wait until ((#Bedroom #MotionSensor).motionsensor_motion == true)	wait until ((#MotionSensor #bedroom).motionsensor_motion == true)
	(#AirConditioner #bedroom).airconditioner_setairconditionermode("auto")
(#Bedroom #AirConditioner).airconditioner_setairconditionermode("auto")	

Row 108 · Category 4

When a smoke is detected in the kitchen, close the kitchen valve.

주방에 연기가 감지되면 주방 밸브를 잠궜줘.

GT name: WhenSmokeIsDetectedInKitchenClose				
Service list: service_list_ver2.0.1.json				
Output name: SmokeDetectedCloseKitchenValve				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9234	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16510
COMPLETION TOKENS 57	TOTAL TOKENS 16567	LLM LATENCY 1.987s	PIPELINE 1.994s	TOK/S 8338.1147
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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SmokeDetector

1 services · 1 values · 0 functions

VALUES

SmokeDetector_Smoke
raw Smoke · arg - · ret BOOL · A Device that detects smoke

FUNCTIONS

None

Valve

3 services · 1 values · 2 functions

VALUES

Valve_ValveState

raw ValveState · arg - · ret BOOL · Controls a valve to open or close it

FUNCTIONS

Valve_Close

raw Close · arg - · ret VOID · Controls a valve to open or close it

Valve_Open

raw Open · arg - · ret VOID · Controls a valve to open or close it

GT JOICODE

cron="" · period=0

```
wait until ((#Kitchen #SmokeDetector).smokedetector_smoke == true)

(#Kitchen #Valve).valve_close()
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#SmokeDetector #Kitchen).smokedetector_smoke == true)
(#Valve #Kitchen).valve_close()
```

FAILURE REASONS

- dataflow
- gt_mismatch

LINE DIFF

GT	Generated
wait until ((#Kitchen #SmokeDetector).smokedetector_smoke == true)	wait until ((#SmokeDetector #Kitchen).smokedetector_smoke == true)
	(#Valve #Kitchen).valve_close()
(#Kitchen #Valve).valve_close()	

Row 109 · Category 4

When the fine dust level in the study reaches 30 or higher, turn on the study air purifier.

서재 미세먼지 농도가 30 이상이 되면 서재 공기청정기를 켜줘.

GT name: WhenFineDustLevelInStudyReaches30				
Service list: service_list_ver2.0.1.json				
Output name: StudyFineDustAirPurifierControl				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.7488	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17525
COMPLETION TOKENS 73	TOTAL TOKENS 17598	LLM LATENCY 2.626s	PIPELINE 2.635s	TOK/S 6701.9575
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT					
Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.					
CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3

AirQualitySensor	CONNECTED
7 services · 7 values · 0 functions	
VALUES	
AirQualitySensor_CarbonDioxide raw CarbonDioxide · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring	
AirQualitySensor_DustLevel raw DustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring	
AirQualitySensor_FineDustLevel raw FineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring	

AirQualitySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Temperature

raw Temperature · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_TvocLevel

raw TvocLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_VeryFineDustLevel

raw VeryFineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

FUNCTIONS

None

AirPurifier

CONNECTED

2 services · 1 values · 1 functions

VALUES

AirPurifier_AirPurifierMode

raw AirPurifierMode · arg - · ret ENUM · Controls air purifier mode

FUNCTIONS

AirPurifier_SetAirPurifierMode

raw SetAirPurifierMode · arg ENUM · ret VOID · Controls air purifier mode

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

GT JOICODE

cron="" · period=0

GENERATED JOICODE

cron="" · period=0

```
wait until ((#Study #AirQualitySensor).airqualitysensor_finedustlevel >= 30)

(#Study #AirPurifier).switch_on()
```

```
wait until ((#AirQualitySensor #study).airqualitysensor_finedustlevel >= 30)
(#AirPurifier #study).airpurifier_setairpurifiermode("auto")
```

FAILURE REASONS

- dataflow
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage

LINE DIFF

GT	Generated
wait until ((#Study #AirQualitySensor).airqualitysensor_finedustlevel >= 30)	wait until ((#AirQualitySensor #study).airqualitysensor_finedustlevel >= 30)
	(#AirPurifier #study).airpurifier_setairpurifiermode("auto")
(#Study #AirPurifier).switch_on()	

Row 110 · Category 4

When a water leak is detected in the basement, sound the main siren in emergency mode.

지하에 누수가 감지되면 메인 사이렌을 비상모드로 울려줘.

GT name: WhenWaterLeakIsDetectedInBasementSound				
Service list: service_list_ver2.0.1.json				
Output name: BasementLeakEmergencySiren				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9633	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16510
COMPLETION TOKENS 62	TOTAL TOKENS 16572	LLM LATENCY 2.613s	PIPELINE 2.620s	TOK/S 6343.1065
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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LeakSensor

1 services · 1 values · 0 functions

VALUES

LeakSensor_Leakage
raw Leakage · arg - · ret BOOL · A Device that senses water leakage

FUNCTIONS

None

CONNECTED

Siren

CONNECTED

2 services · 1 values · 1 functions

VALUES

Siren_SirenMode

raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode

raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

GT JOICODE

cron="" · period=0

```
wait until ((#Basement #LeakSensor).leaksensor_leakage == true)

(#Main #Siren).siren_setsirenmode("emergency")
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#LeakSensor).leaksensor_leakage == true)
(#Siren).siren_setsirenmode("emergency")
```

FAILURE REASONS

- dataflow
- gt_mismatch

LINE DIFF

GT	Generated
wait until ((#Basement #LeakSensor).leaksensor_leakage == true)	wait until ((#LeakSensor).leaksensor_leakage == true)
	(#Siren).siren_setsirenmode("emergency")
(#Main #Siren).siren_setsirenmode("emergency")	

Row 111 · Category 4

When the terrace illuminance falls below 50, turn on the terrace light.

테라스 조도가 50 미만이면 테라스 조명을 켜줘.

GT name: WhenTerraceIlluminanceFallsBelow50TurnOn				
Service list: service_list_ver2.0.1.json				
Output name: TerraceLightOnWhenLowIlluminance				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9150	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 18196
COMPLETION TOKENS 63	TOTAL TOKENS 18259	LLM LATENCY 1.882s	PIPELINE 1.891s	TOK/S 9699.8513
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
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LightSensor

1 services · 1 values · 0 functions

VALUES

LightSensor_Brightness
raw Brightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

None

Light

15 services · 8 values · 7 functions

VALUES

Light_ColorMode

raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness

raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature

raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

GT JOICODE

cron="" · period=0

```
wait until ((#Terrace #LightSensor).lightsensor_brightness < 50)

(#Terrace #Light).switch_on()
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#LightSensor #terrace).lightsensor_brightness < 50)
(#Light #terrace).switch_on()
```

FAILURE REASONS

- dataflow
- gt_mismatch

LINE DIFF

GT	Generated
wait until ((#Terrace #LightSensor).lightsensor_brightness < 50)	wait until ((#LightSensor #terrace).lightsensor_brightness < 50)
	(#Light #terrace).switch_on()
(#Terrace #Light).switch_on()	

Row 112 · Category 4

When the garage door is opened, take a picture with the garage camera.

차고 문이 열리면 차고 카메라로 사진을 찍어줘.

GT name: WhenGarageDoorsOpenedTakePictureWith				
Service list: service_list_ver2.0.1.json				
Output name: GarageDoorOpenTakePicture				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9556	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17153
COMPLETION TOKENS 55	TOTAL TOKENS 17208	LLM LATENCY 2.993s	PIPELINE 3.000s	TOK/S 5749.7995
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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Door

3 services · 1 values · 2 functions

VALUES

Door_DoorState
raw DoorState · arg - · ret ENUM · Allow for the control of a door

FUNCTIONS

Door_Close
raw Close · arg - · ret VOID · Allow for the control of a door

Door_Open

CONNECTED

raw Open · arg - · ret VOID · Allow for the control of a door

Camera

CONNECTED

8 services · 4 values · 4 functions

VALUES

Camera_CameraState

raw CameraState · arg - · ret ENUM · Controls camera device for image/video capture and streaming

Camera_Image

raw Image · arg - · ret BINARY · Controls camera device for image/video capture and streaming

Camera_Stream

raw Stream · arg - · ret STRING · Controls camera device for image/video capture and streaming

Camera_Video

raw Video · arg - · ret BINARY · Controls camera device for image/video capture and streaming

FUNCTIONS

Camera_CaptureImage

raw CaptureImage · arg - · ret BINARY · Controls camera device for image/video capture and streaming

Camera_CaptureVideo

raw CaptureVideo · arg - · ret BINARY · Controls camera device for image/video capture and streaming

Camera_StartStream

raw StartStream · arg - · ret STRING · Controls camera device for image/video capture and streaming

Camera_StopStream

raw StopStream · arg - · ret VOID · Controls camera device for image/video capture and streaming

GT JOICODE

cron="" · period=0

wait until ((#Garage #Door).door_doorstate == "open")

(#Garage #Camera).camera_captureimage()

GENERATED JOICODE

cron="" · period=0

wait until ((#Door).door_doorstate == "open")
(#Camera).camera_captureimage()

FAILURE REASONS

dataflow gt_mismatch

LINE DIFF

GT	Generated
wait until ((#Garage #Door).door_doorstate == "open")	wait until ((#Door).door_doorstate == "open")
	(#Camera).camera_captureimage()
(#Garage #Camera).camera_captureimage()	

Row 113 · Category 4

When the baby room sound exceeds 40 dB, output "Noise detected in the baby room" through the living room speaker.

아기방 소리가 40dB보다 커지면 거실 스피커로 아기방에 소음이 감지되었다고 말해줘.

GT name: WhenBabyRoomSoundExceeds40DBOutputNoise				
Service list: service_list_ver2.0.1.json				
Output name: BabyRoomNoiseAlert				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8650	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17301
COMPLETION TOKENS 60	TOTAL TOKENS 17361	LLM LATENCY 1.911s	PIPELINE 1.919s	TOK/S 9084.7724
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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SoundSensor

1 services · 1 values · 0 functions

VALUES

SoundSensor_Sound

raw Sound · arg · ret DOUBLE · Sound sensor device for measuring sound levels

FUNCTIONS

None

Speaker

CONNECTED

11 services · 2 values · 9 functions

VALUES

Speaker_PlaybackState

raw PlaybackState · arg - · ret ENUM · Speaker device for audio playback and media control

Speaker_Volume

raw Volume · arg - · ret INTEGER · Speaker device for audio playback and media control

FUNCTIONS

Speaker_FastForward

raw FastForward · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Pause

raw Pause · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Play

raw Play · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Rewind

raw Rewind · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_SetVolume

raw SetVolume · arg INTEGER · ret INTEGER · Speaker device for audio playback and media control

Speaker_Speak

raw Speak · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Stop

raw Stop · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_VolumeDown

raw VolumeDown · arg - · ret INTEGER · Speaker device for audio playback and media control

Speaker_VolumeUp

raw VolumeUp · arg - · ret INTEGER · Speaker device for audio playback and media control

GT JOICODE

cron="" · period=0

wait until ((#BabyRoom #SoundSensor).soundsensor_sound > 40)

(#LivingRoom #Speaker).speaker_speak("아기방에 소음이 감지되었습니다.")

GENERATED JOICODE

cron="" · period=0

wait until ((#SoundSensor).soundsensor_sound > 40)
(#Speaker).speaker_speak("Noise detected in the baby room")

FAILURE REASONS

dataflow gt_mismatch

LINE DIFF

GT	Generated
wait until ((#BabyRoom #SoundSensor).soundsensor_sound > 40)	wait until ((#SoundSensor).soundsensor_sound > 40)
	(#Speaker).speaker_speak("Noise detected in the baby room")

GT	Generated
(#LivingRoom #Speaker) .speaker_speak("아기방에 소음이 감지되었습니다.")	

Row 114 · Category 4

When the laundry room dryer's spin speed drops to 5 or below, announce through the living room speaker that the laundry is done.

세탁실 건조기 회전 속도가 5 이하가 되면 거실 스피커로 세탁이 끝났다고 알려줘.

GT name: WhenLaundryRoomDryerSSpinSpeedDropsTo				
Service list: service_list_ver2.0.1.json				
Output name: LaundryDoneAnnouncement				
Retrieval cats: not used (connected_devices scope)				
DET 57.2490	PASS No	GT EXACT No	SIMILARITY 0.3298	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.0000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17709
COMPLETION TOKENS 102	TOTAL TOKENS 17811	LLM LATENCY 2.982s	PIPELINE 2.990s	TOK/S 5971.8357
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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LaundryDryer

4 services · 2 values · 2 functions

VALUES

LaundryDryer_LaundryDryerMode

raw LaundryDryerMode · arg - · ret ENUM · Allows for the control of the laundry dryer mode

LaundryDryer_SpinSpeed

raw SpinSpeed · arg - · ret INTEGER · Allows for the control of the laundry dryer mode

FUNCTIONS

LaundryDryer_SetLaundryDryerMode

raw SetLaundryDryerMode · arg ENUM · ret VOID · Allows for the control of the laundry dryer mode

LaundryDryer_SetSpinSpeed

raw SetSpinSpeed · arg INTEGER · ret VOID · Allows for the control of the laundry dryer mode

Speaker

CONNECTED

11 services · 2 values · 9 functions

VALUES

Speaker_PlaybackState

raw PlaybackState · arg - · ret ENUM · Speaker device for audio playback and media control

Speaker_Volume

raw Volume · arg - · ret INTEGER · Speaker device for audio playback and media control

FUNCTIONS

Speaker_FastForward

raw FastForward · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Pause

raw Pause · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Play

raw Play · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Rewind

raw Rewind · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_SetVolume

raw SetVolume · arg INTEGER · ret INTEGER · Speaker device for audio playback and media control

Speaker_Speak

raw Speak · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Stop

raw Stop · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_VolumeDown

raw VolumeDown · arg - · ret INTEGER · Speaker device for audio playback and media control

Speaker_VolumeUp

raw VolumeUp · arg - · ret INTEGER · Speaker device for audio playback and media control

GT JOICODE

cron="" · period=0

wait until ((#LaundryRoom #LaundryDryer).laundrydryer_spinspeed <= 5)

(#LivingRoom #Speaker).speaker_speak("세탁이 완료되었습니다.")

GENERATED JOICODE

cron="" · period=100

```
triggered := false
spinSpeed = (#LaundryDryer).laundrydryer_spinspeed
if (spinSpeed <= 5) {
  if (triggered == false) {
    (#Speaker).mediaplayback_speak("The laundry is done")
    triggered = true
  }
} else {
```



```
triggered = false
}
```

FAILURE REASONS

- arg_type
- dataflow
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- semantic
- service_match
- unknown_service:mediaplayback_speak

LINE DIFF

GT	Generated
wait until ((#LaundryRoom #LaundryDryer).laundrydryer_spinspeed <= 5)	triggered := false
	spinSpeed = (#LaundryDryer).laundrydryer_spinspeed
(#LivingRoom #Speaker).speaker_speak("세탁이 완료되었습니다.")	if (spinSpeed <= 5) {
	if (triggered == false) {
	(#Speaker).mediaplayback_speak("The laundry is done")
	triggered = true
	}
	} else {
	triggered = false
	}

Row 115 · Category 4

When the temperature in the pantry reaches 25 degrees or higher, turn on the pantry air conditioner.

창고 온도가 25도 이상이 되면 창고 에어컨을 켜줘.

GT name: WhenTemperatureInPantryReaches25DegreesOr				
Service list: service_list_ver2.0.1.json				
Output name: PantryTemperatureACControl				
Retrieval cats: not used (connected_devices scope)				
DET 55.4329	PASS No	GT EXACT No	SIMILARITY 0.7866	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.0000	NUMERIC 0.0000	ENUM 1.0000	PROMPT TOKENS 17045
COMPLETION TOKENS 62	TOTAL TOKENS 17107	LLM LATENCY 2.905s	PIPELINE 2.913s	TOK/S 5888.6097
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
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TemperatureSensor

1 services · 1 values · 0 functions

VALUES

TemperatureSensor_Temperature

raw Temperature · arg - · ret DOUBLE · Get the temperature from a Device that reports current temperature

FUNCTIONS

None

AirConditioner

4 services · 2 values · 2 functions

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

GT JOICODE

cron="" · period=0

```
wait until ((#Pantry #TemperatureSensor).temperaturesensor_temperature >= 25)

(#Pantry #AirConditioner).switch_on()
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#TemperatureSensor #pantry).temperaturemeasurement_temperature >= 25.0)
(#AirConditioner #pantry).switch_on()
```

FAILURE REASONS

- dataflow
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- numeric_grounding
- semantic
- service_match
- unknown_service:temperaturemeasurement_temperature

LINE DIFF

GT	Generated
wait until ((#Pantry #TemperatureSensor).temperaturesensor_temperature >= 25)	wait until ((#TemperatureSensor #pantry).temperaturemeasurement_temperature >= 25.0)
	(#AirConditioner #pantry).switch_on()

GT	Generated
(#Pantry #AirConditioner).switch_on()	

Row 116 · Category 4

When the bathroom humidity exceeds 70%, set the bathroom dehumidifier to freeze protection mode.

욕실 습도가 70%보다 높아지면 욕실 제습기를 동결 방지 모드로 설정해줘.

GT name: WhenBathroomHumidityExceeds70SetBathroomDehumidifi				
Service list: service_list_ver2.0.1.json				
Output name: BathroomHumidityFreezeProtection				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9633	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 0.0000	ENUM 1.0000	PROMPT TOKENS 16611
COMPLETION TOKENS 65	TOTAL TOKENS 16676	LLM LATENCY 2.621s	PIPELINE 2.628s	TOK/S 6362.2143
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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HumiditySensor

1 services · 1 values · 0 functions

VALUES

HumiditySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Allow reading the relative humidity from devices that support it

FUNCTIONS

None

Dehumidifier

CONNECTED

2 services · 1 values · 1 functions

VALUES

Dehumidifier_DehumidifierMode

raw DehumidifierMode · arg - · ret ENUM · Allows for the control of the dehumidifier mode

FUNCTIONS

Dehumidifier_SetDehumidifierMode

raw SetDehumidifierMode · arg ENUM · ret VOID · Allows for the control of the dehumidifier mode

GT JOICODE

cron="" · period=0

```
wait until ((#Bathroom #HumiditySensor).humiditysensor_humidity > 70)

(#Bathroom #Dehumidifier).dehumidifier_setdehumidifiermode("freezeProtection")
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#HumiditySensor).humiditysensor_humidity > 70.0)
(#Dehumidifier).dehumidifier_setdehumidifiermode("freezeProtection")
```

FAILURE REASONS

- dataflow
- gt_mismatch
- numeric_grounding

LINE DIFF

GT	Generated
wait until ((#Bathroom #HumiditySensor).humiditysensor_humidity > 70)	wait until ((#HumiditySensor).humiditysensor_humidity > 70.0)
	(#Dehumidifier).dehumidifier_setdehumidifiermode("freezeProtection")
(#Bathroom #Dehumidifier).dehumidifier_setdehumidifiermode("freezeProtection")	

Row 117 · Category 4

When the office button is pushed, turn on all humidifiers in office.

사무실 버튼이 눌리면 사무실 가습기를 모두 켜줘.

GT name: WhenOfficeButtonIsPushedTurnOnAllHumidifiers				
Service list: service_list_ver2.0.1.json				
Output name: OfficeButtonHumidifierOn				
Retrieval cats: not used (connected_devices scope)				
DET 65.6828	PASS No	GT EXACT No	SIMILARITY 0.6366	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17022
COMPLETION TOKENS 68	TOTAL TOKENS 17090	LLM LATENCY 2.404s	PIPELINE 2.412s	TOK/S 7109.5765
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
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Button

1 services · 1 values · 0 functions

VALUES

Button_Button
raw Button · arg - · ret ENUM · A device with one or more buttons

FUNCTIONS

None

CONNECTED

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

Humidifier

CONNECTED

2 services · 1 values · 1 functions

VALUES

Humidifier_HumidifierMode

raw HumidifierMode · arg - · ret ENUM · Maintains and sets the state of an humidifier

FUNCTIONS

Humidifier_SetHumidifierMode

raw SetHumidifierMode · arg ENUM · ret VOID · Maintains and sets the state of an humidifier

GT JOICODE

cron="" · period=0

wait until ((#Office #Button).button_button == "pushed")

all(#Office #Humidifier).switch_on()

GENERATED JOICODE

cron="" · period=0

wait until ((#Button #office).button_button == "pushed")
(#Humidifier #office).humidifier_sethumidifiermode("auto -")

FAILURE REASONS

dataflow

extraneous

gt_mismatch

gt_receiver_coverage

gt_service_coverage

semantic

LINE DIFF

GT	Generated
wait until ((#Office #Button).button_button == "pushed")	wait until ((#Button #office).button_button == "pushed")
	(#Humidifier #office).humidifier_sethumidifiermode("auto -")
all(#Office #Humidifier).switch_on()	

Row 118 · Category 4

When any presence sensor in the hallway detects presence, set all hallway lights to purple.

복도에 재실 센서가 하나라도 감지되면 모든 복도 조명을 보라색으로 설정해줘.

GT name: WhenAnyPresenceSensorInHallwayDetectsPresenceSet				
Service list: service_list_ver2.0.1.json				
Output name: HallwayPresencePurpleLight				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8270	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 18090
COMPLETION TOKENS 70	TOTAL TOKENS 18160	LLM LATENCY 2.742s	PIPELINE 2.751s	TOK/S 6623.6277
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
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PresenceSensor

1 services · 1 values · 0 functions

VALUES

PresenceSensor_Presence

raw Presence · arg - · ret BOOL · The ability to see the current status of a presence sensor device

FUNCTIONS

None

Light

CONNECTED

CONNECTED

15 services · 8 values · 7 functions

VALUES

Light_ColorMode

raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness

raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature

raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

ColorControl

CONNECTED

2 services · 1 values · 1 functions

VALUES

ColorControl_Color

raw Color · arg - · ret STRING · Allows for control of a color changing device

FUNCTIONS

ColorControl_SetColor
raw SetColor · arg STRING · ret VOID · Allows for control of a color changing device

GT JOICODE

cron="" · period=0

```
wait until (all(#Hallway #PresenceSensor).presencesensor_presence == true)

all(#Hallway #Light).colorcontrol_setcolor("128|0|128")
```

GENERATED JOICODE

cron="" · period=0

```
if (any(#PresenceSensor #hallway).presencesensor_presence == true) {
  all(#Light #hallway).colorcontrol_setcolor("128|0|128")
}
```

FAILURE REASONS

- dataflow
- extraneous
- gt_mismatch

LINE DIFF

GT	Generated
wait until (all(#Hallway #PresenceSensor).presencesensor_presence == true)	if (any(#PresenceSensor #hallway).presencesensor_presence == true) {
	all(#Light #hallway).colorcontrol_setcolor("128 0 128")
all(#Hallway #Light).colorcontrol_setcolor("128 0 128")	}

Row 119 · Category 4

When the carbon monoxide level in the server room becomes 5000 ppm or above, sound all sirens in emergency mode.

서버실 일산화탄소 농도가 5000 ppm 이상이되면 모든 사이렌을 비상모드로 울려줘.

GT name: WhenCarbonMonoxideLevelInServerRoomBecomes				
Service list: service_list_ver2.0.1.json				
Output name: ServerRoom_CO_EmergencySiren				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8958	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17187
COMPLETION TOKENS 70	TOTAL TOKENS 17257	LLM LATENCY 4.434s	PIPELINE 4.443s	TOK/S 3891.6201
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 4	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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AirQualitySensor

7 services · 7 values · 0 functions

VALUES

AirQualitySensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_DustLevel

raw DustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_FineDustLevel

raw FineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

CONNECTED

AirQualitySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Temperature

raw Temperature · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_TvocLevel

raw TvocLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_VeryFineDustLevel

raw VeryFineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

FUNCTIONS

None

Siren

CONNECTED

2 services · 1 values · 1 functions

VALUES

Siren_SirenMode

raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode

raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

GT JOICODE

cron="" · period=0

```
wait until ((#ServerRoom #AirQualitySensor).airqualitysensor_carbondioxide >= 5000)

all(#Siren).siren_setsirenmode("emergency")
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#AirQualitySensor).airqualitysensor_carbondioxide >= 5000)
(#Siren).siren_setsirenmode("emergency")
```

FAILURE REASONS

- dataflow
- gt_mismatch
- semantic

LINE DIFF

GT	Generated
wait until ((#ServerRoom #AirQualitySensor).airqualitysensor_carbondioxide >= 5000)	wait until ((#AirQualitySensor).airqualitysensor_carbondioxide >= 5000)
	(#Siren).siren_setsirenmode("emergency")
all(#Siren).siren_setsirenmode("emergency")	

Row 120 · Category 4

When the bedroom shade button is pushed, lower the shade.

안방 셰이드 버튼이 눌리면 셰이드를 내려줘.

GT name: WhenBedroomShadeButtonIsPushedLowerShade				
Service list: service_list_ver2.0.1.json				
Output name: BedroomShadeButtonLowerShade				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9000	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17035
COMPLETION TOKENS 65	TOTAL TOKENS 17100	LLM LATENCY 2.121s	PIPELINE 2.128s	TOK/S 8061.8547
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
---	--	--	------------------------	----------------------	--------------------------

Button

1 services · 1 values · 0 functions

VALUES

Button_Button
raw Button · arg - · ret ENUM · A device with one or more buttons

FUNCTIONS

None

CONNECTED

WindowCovering

CONNECTED

6 services · 2 values · 4 functions

VALUES

WindowCovering_CurrentPosition

raw CurrentPosition · arg - · ret INTEGER · Controls a window covering to open or close it

WindowCovering_WindowCoveringType

raw WindowCoveringType · arg - · ret ENUM · Controls a window covering to open or close it

FUNCTIONS

WindowCovering_DownOrClose

raw DownOrClose · arg - · ret VOID · Controls a window covering to open or close it

WindowCovering_SetLevel

raw SetLevel · arg INTEGER · ret INTEGER · Controls a window covering to open or close it

WindowCovering_Stop

raw Stop · arg - · ret VOID · Controls a window covering to open or close it

WindowCovering_UpOrOpen

raw UpOrOpen · arg - · ret VOID · Controls a window covering to open or close it

GT JOICODE

cron="" · period=0

wait until ((#Bedroom #Shade #Button).button_button == "pushed")

(#Bedroom #Shade).windowcovering_downorclose()

GENERATED JOICODE

cron="" · period=0

wait until ((#Button #bedroom).button_button == "pushed")
(#WindowCovering #bedroom).windowcovering_downorclose()

FAILURE REASONS

- dataflow
- extraneous
- gt_mismatch
- semantic

LINE DIFF

GT	Generated
wait until ((#Bedroom #Shade #Button).button_button == "pushed")	wait until ((#Button #bedroom).button_button == "pushed")
	(#WindowCovering #bedroom).windowcovering_downorclose()
(#Bedroom #Shade).windowcovering_downorclose()	